NEUROADHD

# Understanding the Model Dashboard

A plain-English guide to every page, the numbers it shows, and what those numbers do - and do not - mean.

**What this website is**
A research dashboard for viewing ADHD-200 model benchmark results, the data split, selected connectivity features, and model explanations. It is not a medical test and does not accept MRI uploads for diagnosis.

## Website reviewed

NeuroADHD_ModelDashboard - a React and Vite frontend. The visible navigation contains nine pages. The frontend is currently open at **http://127.0.0.1:5175/**.

## Who this guide is for

Readers who want to understand what each page means without needing machine-learning or neuroimaging experience.

| ADHD-200 CC200 time courses | Pairwise ROI correlations | 19,900 edge features | Select 100 features | Classifier and evaluation |
|---|---|---|---|---|

The pages describe this workflow and show a saved set of benchmark results. The visible frontend reads a local data snapshot; it is not a live patient prediction application.

START HERE

# What you are looking at

This section explains the cohort labels and the dashboard frame before the page-by-page tour.

The study uses resting-state fMRI from ADHD-200. A scan is represented as activity over time in 200 CC200 regions. The project computes correlations between pairs of regions, giving 19,900 possible connections per subject. The model uses a selected set of 100 connections.

The headline cohort numbers are:

| Partition | People | ADHD | Control | Site assignment |
|---|---|---|---|---|
| Train | 114 | 30 | 84 | Six sites: KKI, NYU, OHSU, Peking_1, Peking_3, Pittsburgh |
| Validation | 19 | 10 | 9 | NeuroIMAGE |
| Test | 20 | 10 | 10 | Peking_2 |

The sidebar says 7 development sites and 1 locked test site: six sites are in train, NeuroIMAGE is the seventh development site, and Peking_2 is the separate test site.

## Shared parts of every page

- **Research-only badge and warning:** results are experimental and are not medical diagnoses.
- **Top badges:** identify the selected model setting and Peking_2 test cohort.
- **Sidebar:** choose one of the nine visible pages. The page changes without changing the browser URL.

## How the visible frontend works

The active React app imports benchmark data from a bundled JavaScript file. It does not request the page data from the FastAPI backend. That means the dashboard can display its saved snapshot even if the backend is stopped, but changes to benchmark results will not appear automatically.

**Useful distinction**
This dashboard displays analyses of a fixed dataset. It does not contain an MRI upload form or an individual patient diagnosis workflow.

# Overview

A quick dashboard summary of the data, selected model, and headline evaluation metrics.

The cards state: ADHD-200 is the dataset; CC200 is the 200-region brain parcellation; 153 subjects are included; the original connectivity table has 19,900 edges; 100 edges were selected; and the chosen classifier is described as a Balanced Random Forest.

The split table shows who belongs to train, validation, and test. “Development” means data used while building or comparing models. The Peking_2 test site is supposed to be the final independent check.

## The two summary scores shown are:

| 0.6045 | 0.6026 | 0.5600 |
|---|---|---|
| Mean ROC-AUC across 8 CV repetitions | Mean balanced accuracy across 8 CV repetitions | ROC-AUC on Peking_2, 20 people |

A score of 0.50 ROC-AUC is roughly chance-level ranking. These dashboard scores are only a little higher than 0.50, so they should be read as modest results, especially given the small number of subjects.

**Read this page as**
“Here is the cohort, the model we chose, and a summary of saved experiments.” It is not proof that the model diagnoses ADHD reliably.

# Model & Pipeline

Shows how the time-series files are converted into model inputs and lists the chosen model settings.

The pipeline is displayed in this order:

| Step | Plain-English meaning |
|---|---|
| 1. CC200 time courses | For each person, values over time in 200 functional brain regions. |
| 2. Pearson correlations | For each pair of regions, measure how similarly their signals change. |
| 3. Fisher z | Convert correlation values to a scale useful for statistical modeling. |
| 4. Median imputation | If an edge value is missing, fill it using the median learned from training data. |
| 5. Zero-variance filter | Drop features that are constant in the training data. |
| 6. Standard scaling | Put feature values on comparable scales using training-only statistics. |
| 7. ANOVA SelectKBest | Rank edges by a univariate statistical score and keep 100, fitted inside training folds. |
| 8. Random Forest | Combine many decision trees to predict the class. |

The page lists 50 trees, maximum depth 3, square-root feature sampling, class_weight=balanced, and random seed 42. “Class weight balanced” makes errors on the less common class count more heavily during training.

**Naming caveat**
The dashboard calls this “Balanced Random Forest,” while its metadata names sklearn RandomForestClassifier with class_weight="balanced". Those are not necessarily the same estimator. The source metadata is more specific; verify the exact implementation before reporting the model name.

# Robustness & Test

Shows repeated cross-validation results and the reported one-time Peking_2 test results.

## Block A: repeated cross-validation

The page lists five-fold stratified cross-validation repeated with eight random seeds (42 to 49). Each repetition predicts each development subject using a fold where that subject was held out. The page reports the mean and spread across repetitions.

| Metric | Plain-English meaning | Displayed summary |
|---|---|---|
| ROC-AUC | How well the model ranks ADHD cases above controls across thresholds. 0.50 is chance-like. | 0.6045 +/- 0.0163 |
| Accuracy | Share of all predictions that were correct. | 0.6447 |
| Balanced accuracy | Average of ADHD sensitivity and control specificity. | 0.6026 |
| Sensitivity | Share of ADHD subjects predicted ADHD. | 0.4969 |
| Specificity | Share of controls predicted control. | 0.7083 |
| F1 | Combines precision and sensitivity into one score. | 0.4566 |

## Block B: Peking_2 test

The page reports accuracy 0.60, balanced accuracy 0.60, sensitivity 0.60, specificity 0.60, F1 0.60, ROC-AUC 0.56, and PR-AUC 0.6341. On only 20 people, this corresponds to 12 correct and 8 incorrect classifications.

| | Predicted ADHD | Predicted Control |
|---|---|---|
| Actual ADHD | 6 (true positive) | 4 (false negative) |
| Actual Control | 4 (false positive) | 6 (true negative) |

PR-AUC summarizes precision versus sensitivity as the classification threshold changes. Its value depends on how common the positive class is, so compare it only under the same cohort and protocol.

# Hybrid Comparison

Compares a single Random Forest with voting and stacking ensembles.

A voting model combines predictions from multiple classifiers. Soft voting averages their probability scores; hard voting combines their class votes. Stacking trains a second model to combine the outputs of base models.

The page's displayed comparison is:

| Approach | CV repetitions shown | ROC-AUC mean | Balanced accuracy | Sensitivity |
|---|---|---|---|---|
| Balanced Random Forest | 8 x 5-fold | 0.6045 | 0.6026 | 0.4969 |
| Soft voting: 60% RF / 40% XGBoost | 5 x 5-fold | 0.5926 | 0.5798 | 0.4500 |
| Hard voting: RF + XGBoost | 5 x 5-fold | 0.5926 | 0.5798 | 0.4500 |
| Stacking with logistic regression | 5 x 5-fold | 0.5555 | 0.5428 | 0.5050 |

The page concludes that the hybrid models did not beat the single Random Forest on its shown average ROC-AUC.

**Comparison caveat**
The page says the protocols are identical, but it lists 8 repetitions for the Random Forest and 5 for the hybrids. Hard and soft voting also show identical summary numbers. Treat the ranking as a summary to investigate, not a perfectly matched head-to-head test.

# SHAP Explainability

Ranks connections and regions that influenced the fitted model.

Each feature is a connection between two CC200 regions, such as ROI_023__ROI_189. Mean absolute SHAP is the average size of the feature's contribution across the subjects included in the analysis. Bigger values mean the feature mattered more to predictions on average; absolute SHAP does not tell the direction by itself.

The page also shows a permutation score. This asks how model performance changes when a feature is shuffled. A larger drop suggests the model relied more on that feature in the evaluated data.

The ROI hub summary counts how often each region appears among important connections. The table can be searched and sorted. Anatomical region names are intentionally not supplied because CC200 node IDs are functional clusters, not direct anatomical labels.

**How to interpret it**
A SHAP result explains the model's behavior for these data. It does not prove that an edge causes ADHD or is a biological marker. Multi-site scanner differences and small sample size can also affect the ranking.

# Connectivity Network

Turns the important connectivity edges into a clickable graph.

Each circle is a CC200 node (one of the 200 functional clusters). Lines connect node pairs that appear among the selected important edges. The graph highlights a small subnetwork rather than all 19,900 possible connections.

The legend distinguishes higher-importance hub nodes, other selected nodes, and edges associated with a higher ADHD model score. Click a node or line to see its feature and attribution details.

The graph is a view of model-selected features, not a full anatomical brain map. It does not tell you where a cluster sits anatomically unless a valid atlas mapping is available.

**Keep in mind**
The graph is a visualization of statistical model inputs. It is not a scan, brain activation image, or causal map.

# Subject Decompositions

Shows individual example predictions and which edges pushed the prediction up or down.

Choose one of the listed subjects. The page shows its predicted ADHD probability, true recorded diagnosis, predicted class, site, and the strongest positive and negative SHAP contributions.

A positive contribution moves this model’s output toward ADHD; a negative contribution moves it toward Control. The displayed z value is the feature value after the model’s preprocessing, not a clinical measurement or probability.

Examples include correct ADHD predictions, a near-boundary case, a correct Control prediction, a false positive, and Peking_2 test cases. This helps explain individual model decisions, but a few examples do not summarize overall performance.

**Important test-set note**
This page displays individual Peking_2 outcomes. Combined with earlier completed test-evaluation runs found in the benchmark files, the Peking_2 set should not be described as still unseen or untouched.

# Methodology & Splits

Explains which site groups were assigned to model development and testing.

The intended site-based split keeps all people from the same site together: six sites for training, NeuroIMAGE for validation, and Peking_2 for the final test. This is useful because scanners, acquisition settings, and participant populations can differ by site.

The page describes preprocessing as fold-local: medians, scaling values, and ANOVA edge rankings should be learned from training subjects in each fold, then applied to held-out subjects. That is the right general rule for avoiding preprocessing leakage.

## What the benchmark code shows

The repeated-RF and repeated-hybrid scripts use StratifiedKFold on a 133-person development dataset (train plus validation). StratifiedKFold balances labels but does not keep sites intact across folds. So the 8-repeat CV score is a subject-level estimate within mixed sites, not a separate-site CV estimate. The Peking_2 evaluation is the separate-site check.

**Do not overread the “locked” label**
The stored benchmark history includes at least two earlier completed run reports with final_test_executed=true before the later run shown on the dashboard. The current Peking_2 score is therefore best treated as exploratory; it is not a one-time untouched holdout anymore.

# Limitations

The dashboard’s own page lists the main boundaries of the study.

- **Small cohort:** 153 subjects is too small to support clinical conclusions, and each test subject has a large effect on the reported score.
- **Different sites:** scanners, acquisition protocols, age groups, and demographics may differ. A model can learn site patterns as well as diagnostic patterns.
- **CC200 clusters:** ROI numbers are data-driven functional regions, not a direct one-to-one anatomical naming system.
- **Explanations are not causes:** SHAP and permutation importance describe model reliance, not biological mechanisms.
- **Not clinically validated:** the system is for academic benchmarking only, not diagnosis or treatment planning.

**Plain-language takeaway**
This dashboard helps researchers inspect a model experiment. Its scores are modest, based on a small and multi-site dataset, and the displayed test set has already been evaluated in earlier runs. Do not use it to diagnose a person.

EXTRA DETAILS

# Pages in the code that are not on the sidebar

The frontend contains additional view components, but the current App does not connect them to its nine-page navigation.

| Component | What it was designed to do |
|---|---|
| DashboardView | System summary, live validation leaderboard, and buttons to run one or all models. |
| ModelsView | Pick a classifier and inspect its configuration; can start a model run. |
| ComparisonView | Compare validation metrics across classifiers. |
| EnsembleView | Show ensemble and hybrid model results. |
| FinalTestView | Display or trigger the Peking_2 final evaluation. |
| ModelHistoryView | Request model performance history from the backend API. |
| RunsHistoryView / RunsView | Browse saved runs and audit records. |
| RunCompareView | Compare selected benchmark runs via the backend. |
| RunDetailModal | Detailed run tabs for data, preprocessing, models, test, environment, and AI summary. |
| ReportView | Embed a generated HTML report. |

The visible frontend page does not use these components. The FastAPI backend includes model-run endpoints, but it was not started during this walkthrough and no model was run.

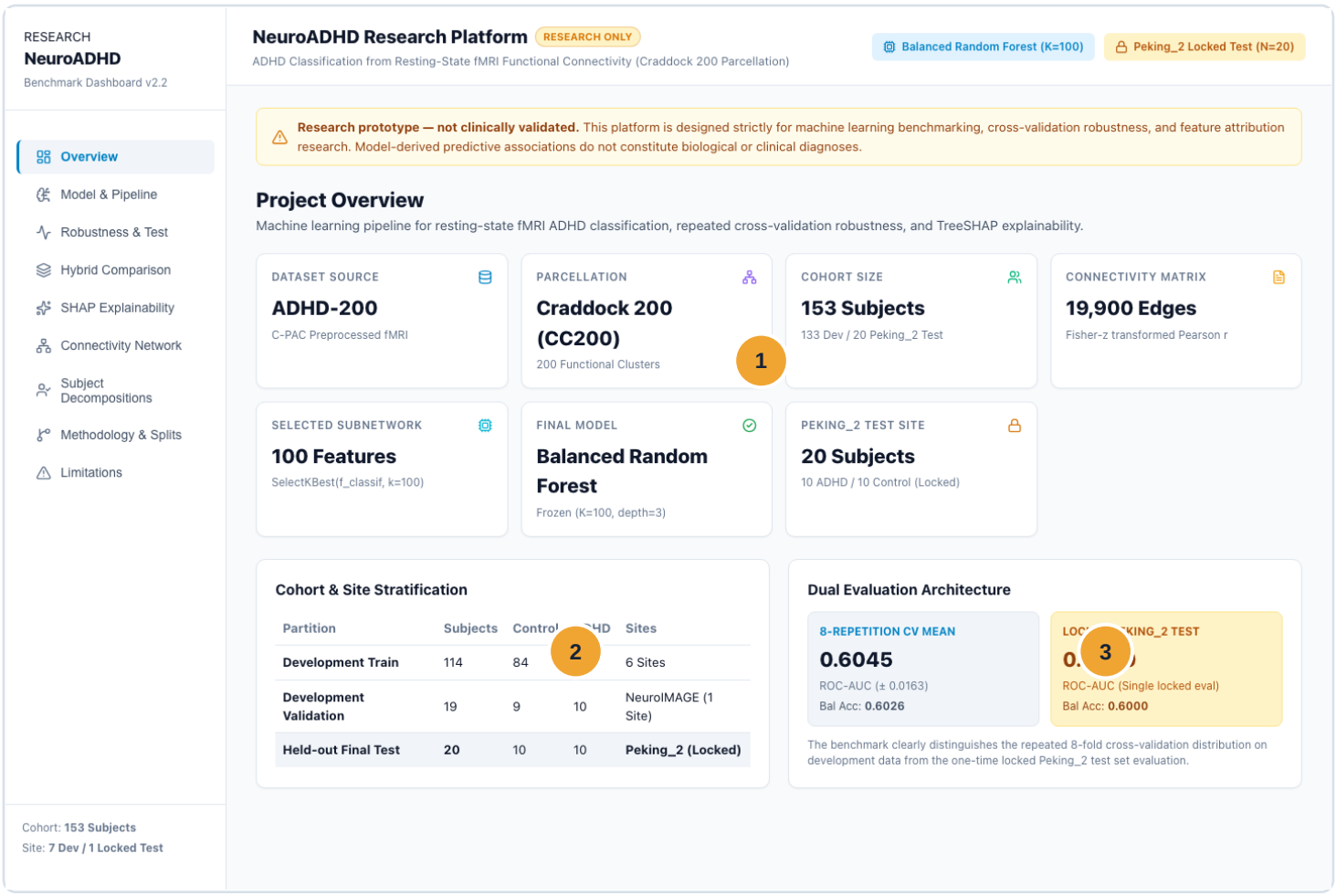
## Glossary

| Term | Meaning |
|---|---|
| ADHD / Control | The dataset labels. They are not generated by this dashboard. |
| CC200 / ROI | 200 functional brain regions and a region’s time-course signal. |
| Edge / connectivity feature | A correlation between two ROIs. There are 200 x 199 / 2 = 19,900 unique pairs. |
| Fisher z | A transformation of correlation r, commonly written atanh(r), to make values more suitable for statistical analysis. |
| Cross-validation | Repeat train/holdout folds to estimate performance using available development subjects. |
| ROC-AUC | How well higher model scores rank ADHD above Control across possible thresholds. 0.5 is chance-like. |
| Sensitivity / specificity | ADHD detection rate / Control detection rate. |
| SHAP | A method for describing how features contributed to a model output. |
| Leakage | Information from held-out data accidentally influencing learned preprocessing or model choices. |

Sources inspected: the active React app and its nine view files, bundled benchmark data, the data-compilation script, the benchmark CV scripts, and saved run reports. Raw scans were not read for this guide.

# Overview

A quick summary of the dataset, model, subject groups, and reported results.




## 1 Summary cards

These cards tell you what data and model the dashboard uses, how many people are included, and how many connectivity features were kept.

## 2 Who is in each group?

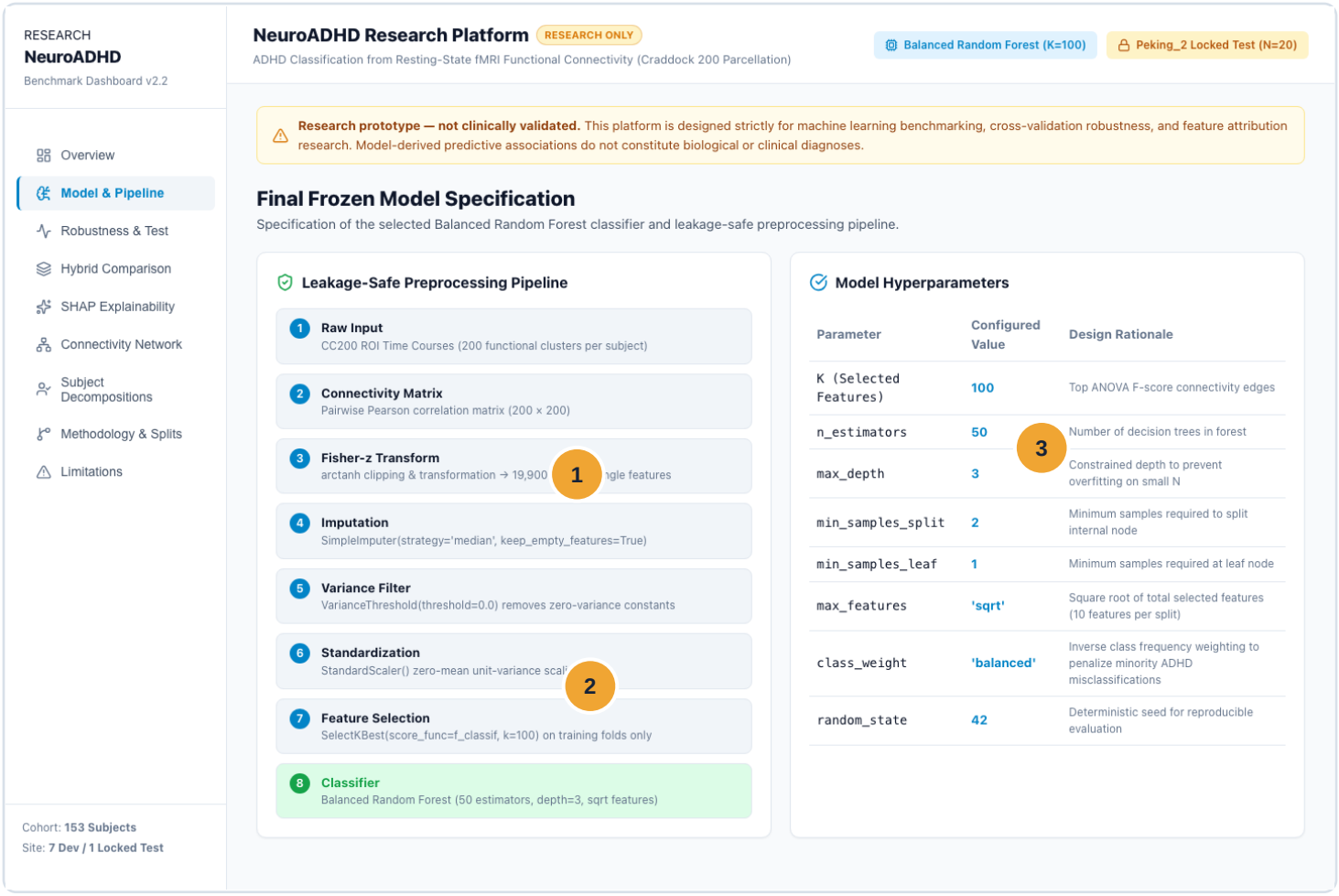
This table shows the numbers of Control and ADHD subjects in training, validation, and the held-out test group.

## 3 How well did it do?

These scores summarize cross-validation and the Peking_2 test. They measure this experiment, not a clinical diagnosis.

# Model & Pipeline

Shows the steps that turn brain scans into the final model prediction.




### 1 From scans to inputs

The left column follows the data from CC200 time courses to 19,900 pairwise connectivity values.

### 2 Preparing the data

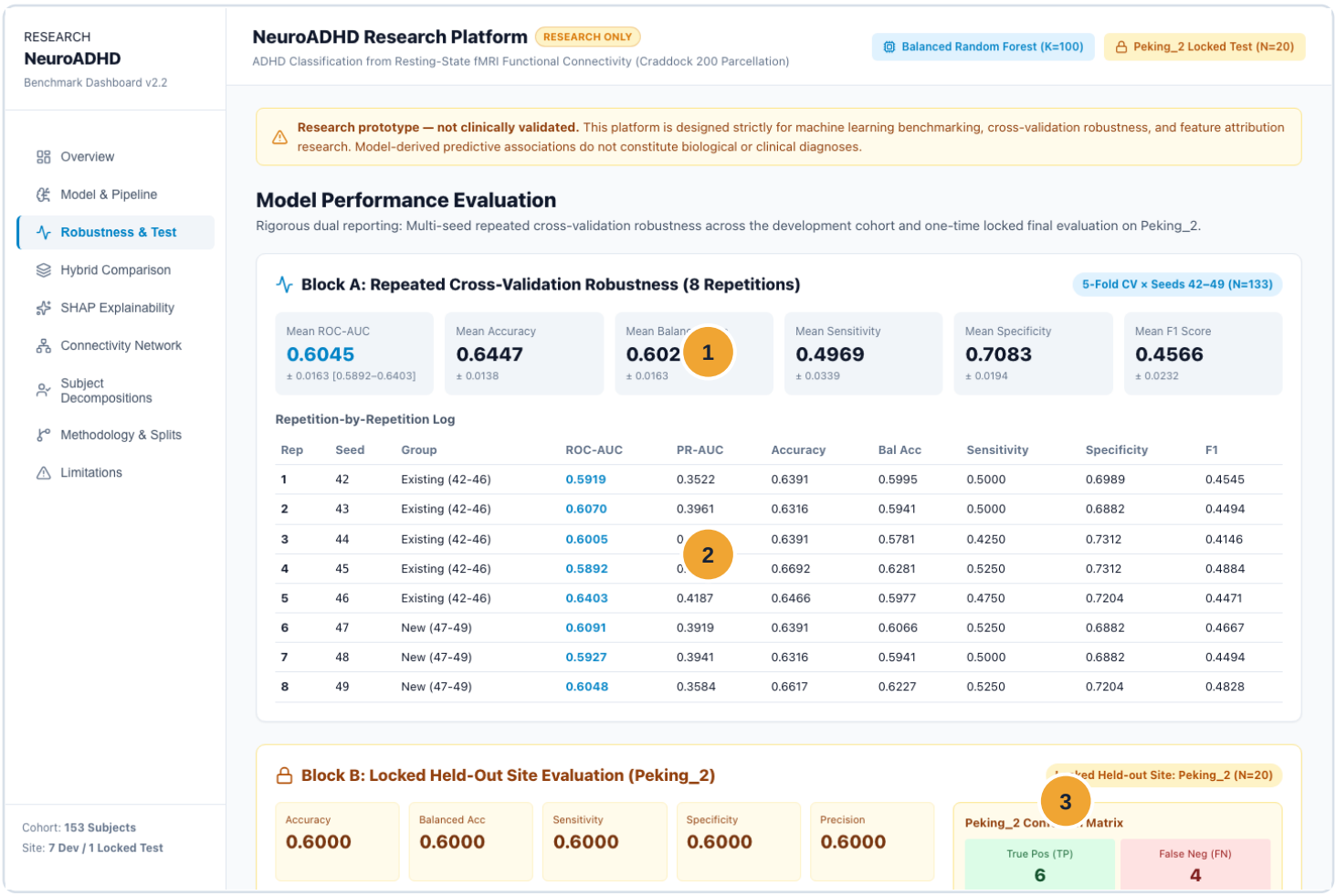
Imputation, scaling, and choosing 100 features happen inside training folds so held-out data does not set those values.

### 3 Model settings

The right table lists the Random Forest settings used for the reported model.

# Robustness & Test

Shows repeated development checks and a separate held-out site result.




### 1 Repeated checks

The top cards summarize five-fold cross-validation repeated with eight random seeds on the development group.

### 2 Run-by-run results

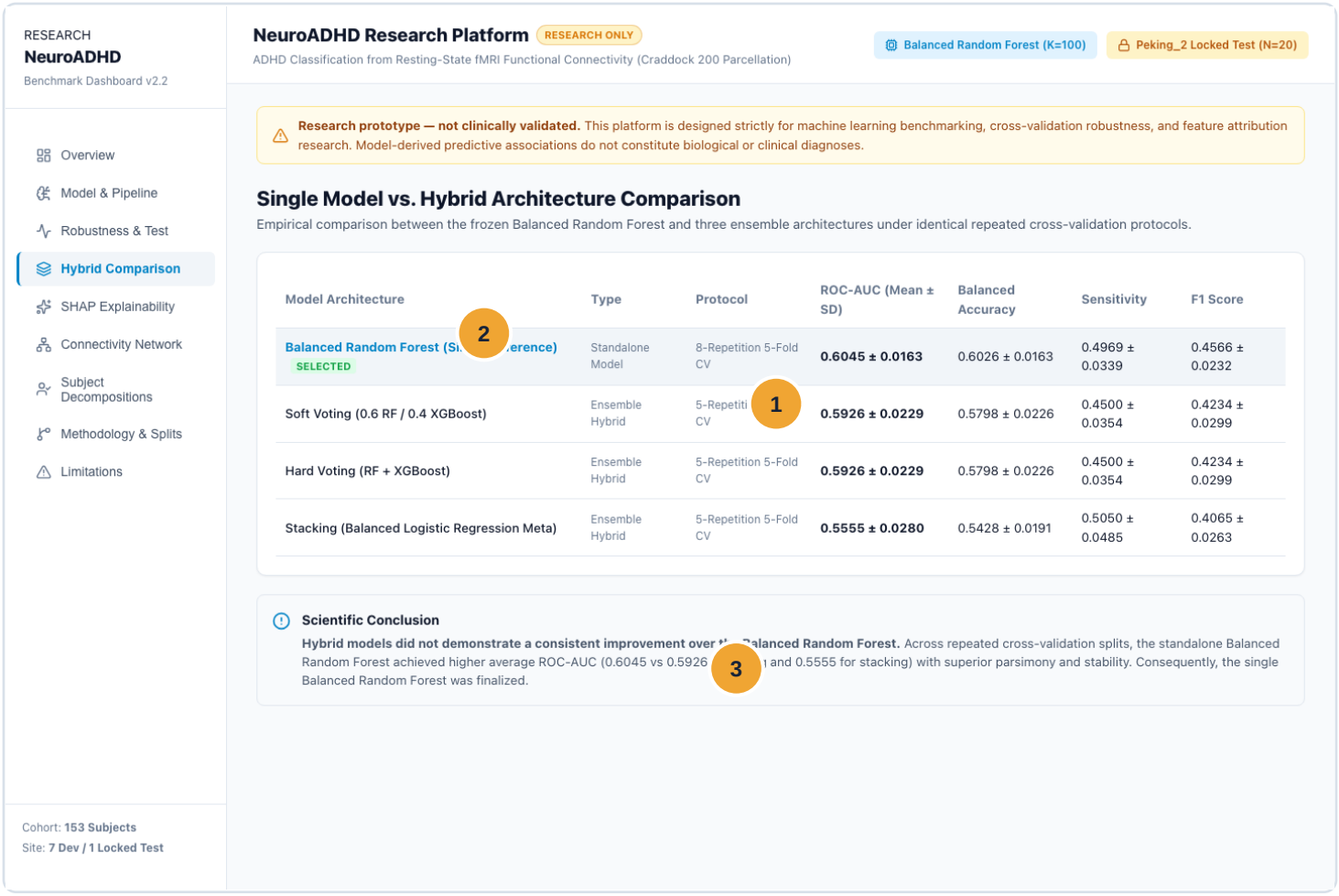
Each row is one random seed. The spread helps show how much scores change when the folds change.

### 3 Held-out site

The lower section reports predictions for Peking_2. The confusion matrix counts correct and incorrect ADHD/Control predictions.

# Hybrid Comparison

Compares the single Random Forest with three models that combine predictions.




### 1 Model comparison

Each row is a model approach. The columns compare ROC-AUC, balanced accuracy, sensitivity, and F1.

### 2 Selected model

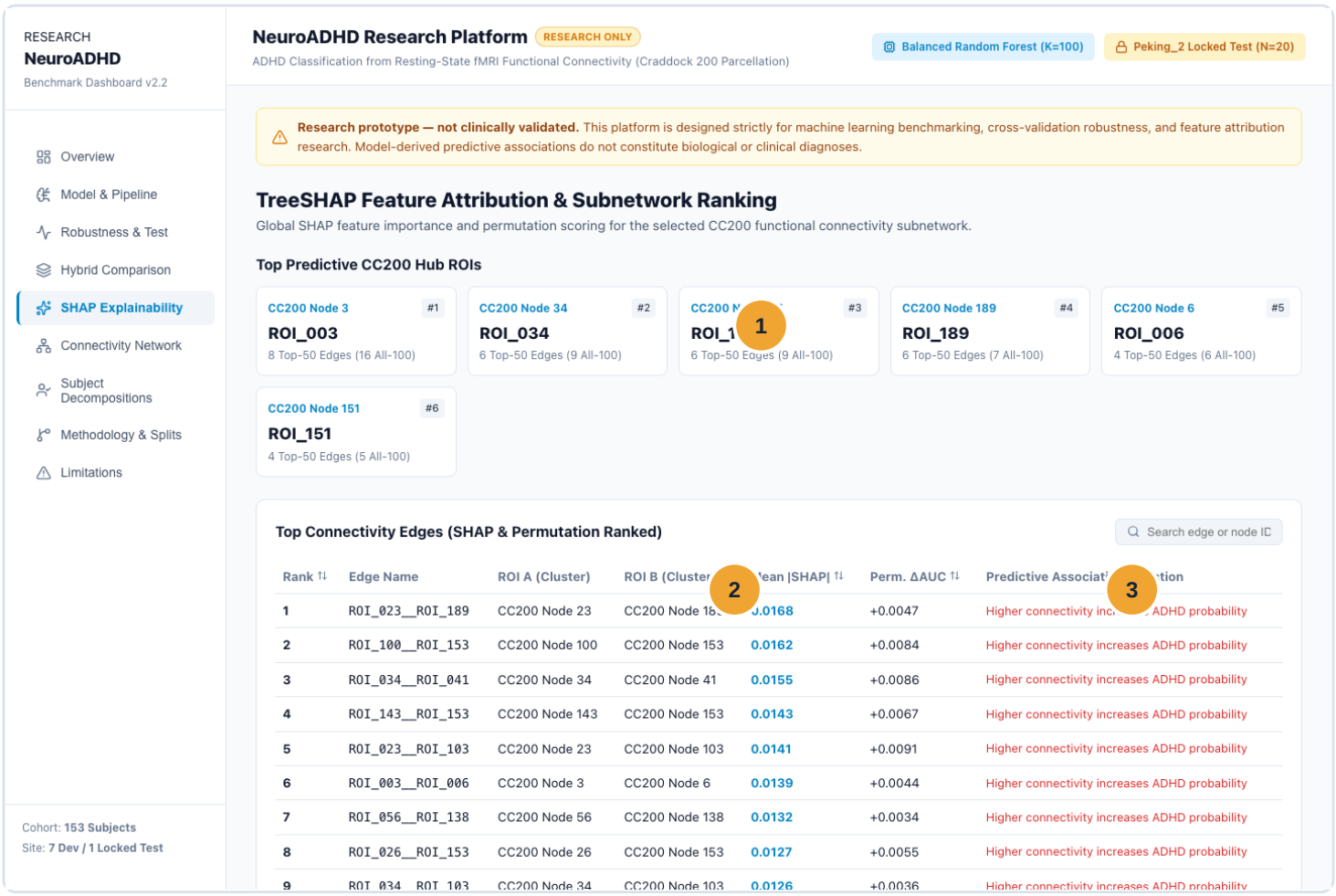
The shaded row marks the single Random Forest chosen in this dashboard.

### 3 Conclusion

This note says the combined models did not score better in the reported cross-validation comparison.

# SHAP Explainability

Ranks connectivity values that influenced the model's predictions.




### 1 Frequently involved regions

These cards show CC200 node numbers that appear often among the highly ranked edges.

### 2 Important connections

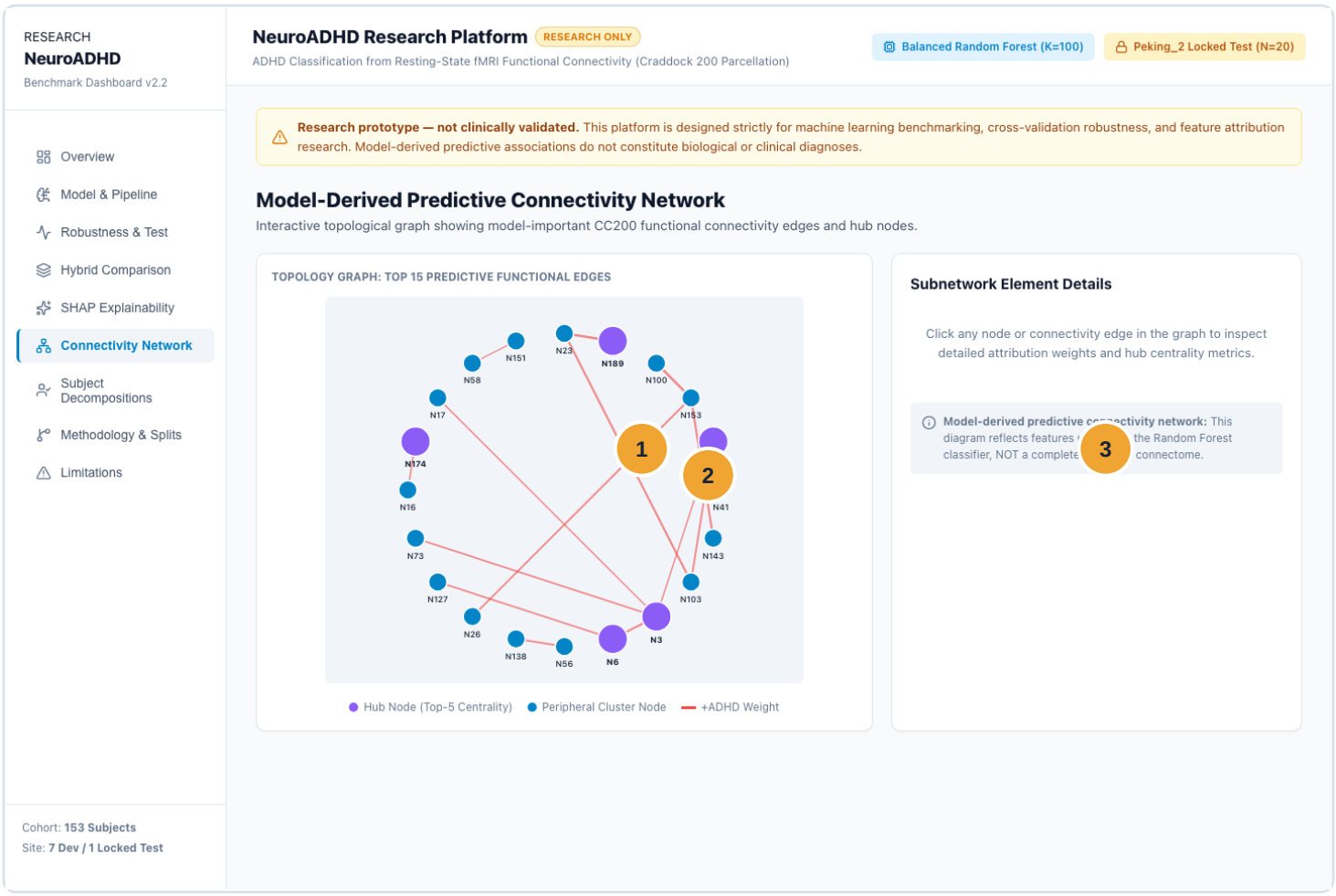
Each table row is a pair of regions. Mean absolute SHAP summarizes influence size; permutation change shows how a score shifts when an edge is shuffled.

### 3 Direction

The last column describes the direction associated with the model output. This is a model pattern, not proof that a brain connection causes ADHD.

# Connectivity Network

Draws the highly ranked connections as a small network.




### 1 Dots are CC200 regions

Each labeled dot is one data-driven CC200 region. Larger purple dots mark the hubs emphasized by the dashboard.

### 2 Lines are model features

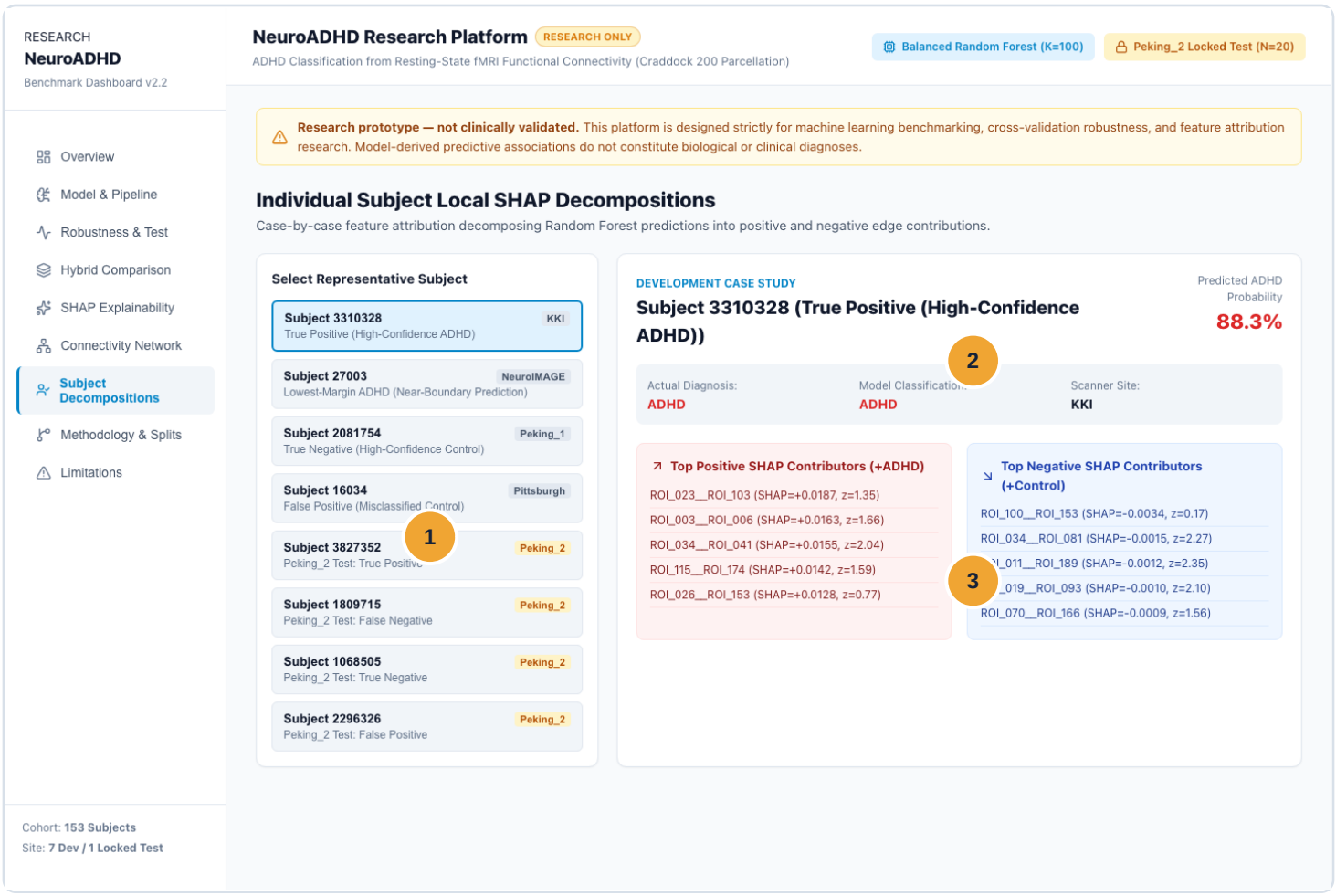
The lines connect region pairs used as predictive features; color and thickness indicate the displayed ADHD-associated weight.

### 3 Details panel

Selecting a node or line is intended to show its attribution details here. The graph is only the selected predictive subnetwork, not the whole brain network.

# Subject Decompositions

Explains one example prediction at a time.




### 1 Choose an example

Pick a subject on the left. The list includes correct predictions, a near-boundary case, and errors.

### 2 Read the prediction

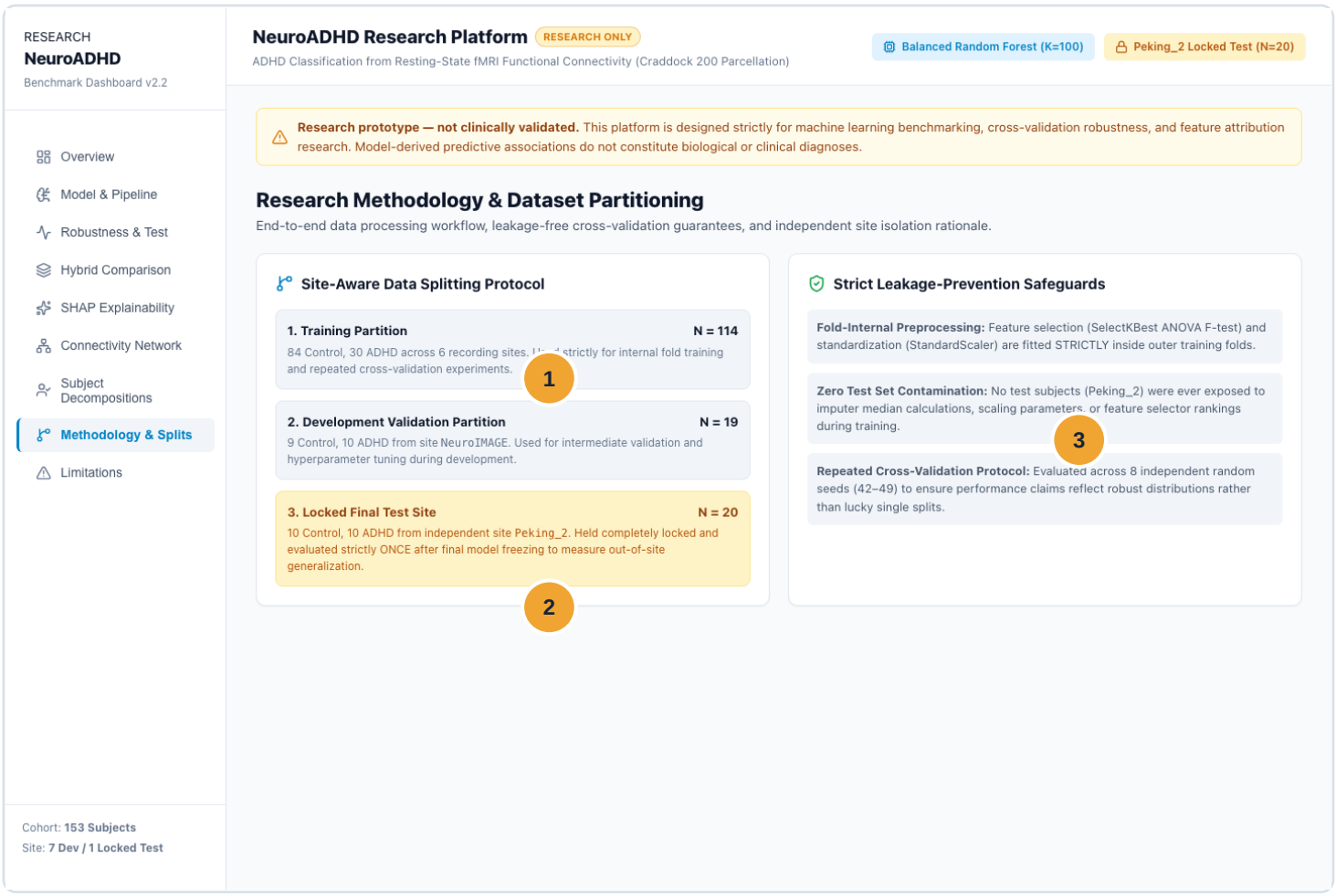
The panel shows the model's ADHD probability, the recorded diagnosis, the predicted class, and the site.

### 3 Positive and negative clues

Positive SHAP values pushed this prediction toward ADHD; negative values pushed it toward Control. They explain this model output only.

# Methodology & Splits

Explains how subjects were separated for development and final evaluation.




### 1 Training group

The model is developed using 114 subjects from six sites.

### 2 Validation and held-out test

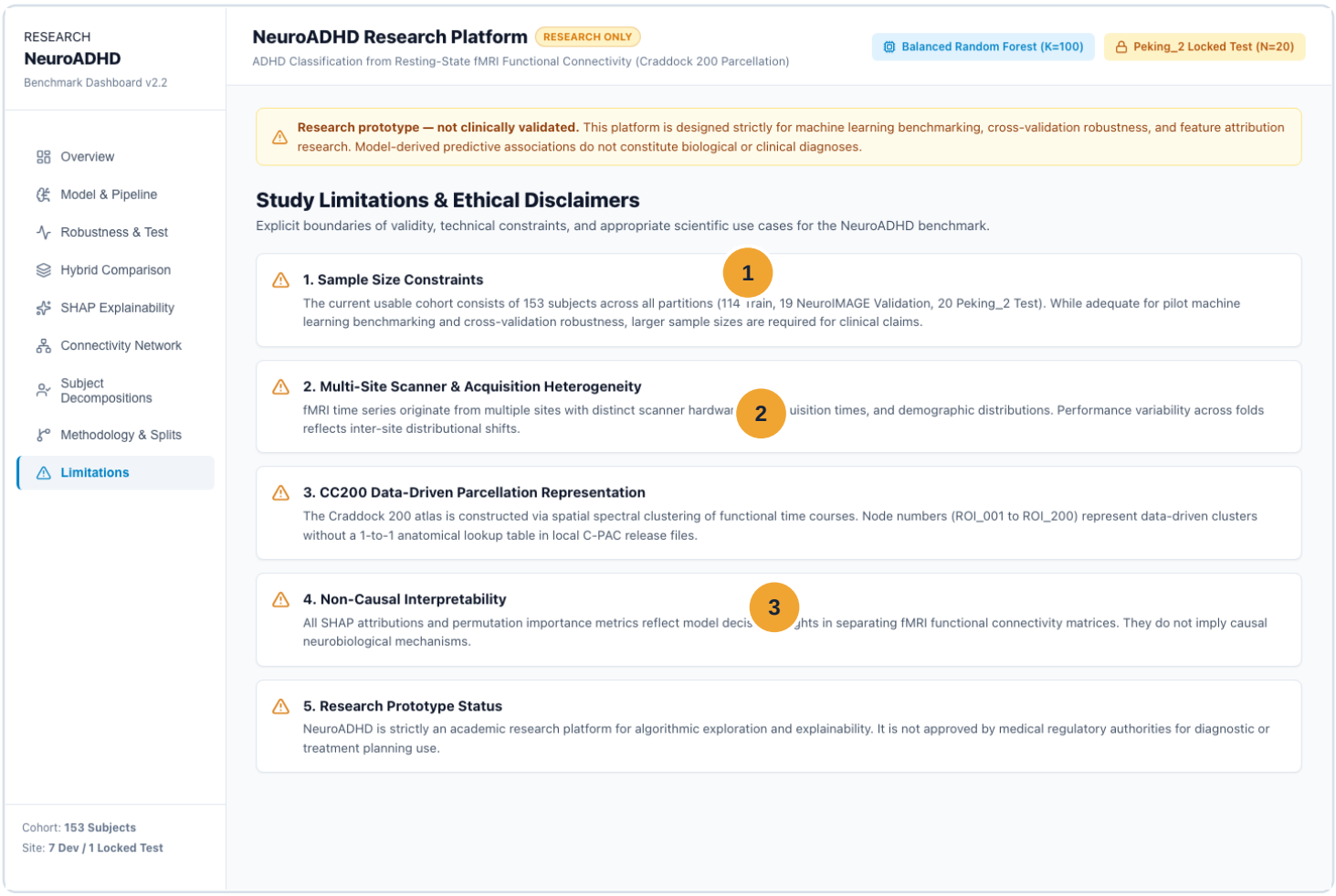
Nineteen NeuroIMAGE subjects are for development checks. Twenty Peking_2 subjects form the separate held-out site.

### 3 Leakage safeguards

The panel summarizes that learned preprocessing is fit within training folds and the test group is kept out of model fitting.

# Limitations

Lists reasons to treat the results as early research evidence.




1

## Small sample

153 people are enough for a pilot benchmark, but not enough to support clinical claims.

2

## Different scanners and sites

Scanner and acquisition differences can affect fMRI measurements and model performance.

3

## Interpretation and intended use

CC200 nodes are data-driven clusters, and model explanations are not causal evidence. The dashboard is a research prototype, not a diagnostic tool.